# Equivariant observers: applications to autonomous systems

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Abstract: The operation of all autonomous systems depends critically on their ability to estimate their dynamic state. For high performance small-scale aerial, marine and terrestrial systems with limited sensor suites, highly dynamic motion, and limited computational capacity, the state observer performance is even more important. The role of symmetry as the fundamental structure needed to design the state-of-the-art observers and filters for autonomous systems has become accepted over the last decade and having an understanding is critical for student planning to work with the latest technology.

This course provides an introduction to the modern geometric theory underlying design of observers and filters for autonomous systems. The course provides a strong introduction to the tools of matrix calculus and matrix Lie groups from an engineering perspective. The modern theory of observer and filter design is taught through an extensive suite of case studies drawn from autonomous systems applications including; attitude estimation, velocity aided attitude estimation, inertial navigation systems, visual inertial odometry, and homography estimation. The course takes a hands on approach and students will learn the tools and techniques to derive and implement nonlinear observers and filters for real world robotic systems.

## Topics:

1) Perspectives on observer and filter design for physical systems.
2) Matrix calculus and matrix ODEs.
3) Lyapunov observer design for systems with matrix Lie group state.
4) Numerical implementation, measurement bias, and time delays.
5) Lie theory foundations, Kinematic systems with symmetry.
6) General theory of observer and filter design.
7) Overview of recent symmetry structures.

Algorithms for attitude estimation, homography estimation, velocity aided attitude estimation, inertial navigation systems (INS), and visual inertial odometry (VIO) will be covered in the course. Practical work in this course uses MATLAB (or equivalent scripting language such as Python) extensively. Students are required to have a working system on their own laptop for the course.